

Java Inheritance and Interface Relationships

A worked reference guide with programs, sample output, and test-case verification

1. C1 extends C2

Aim

To demonstrate single inheritance in Java using the extends keyword.

Definition

Single inheritance occurs when one class inherits the properties and methods of another class. Here, class C1 inherits from class C2.

Syntax

```
class C1 extends C2 {  
    // body of C1  
}
```

Structure

```
      C2  (Parent)  
      |  
extends  
      |  
      v  
      C1  (Child)
```

Program

```
class C2 {  
    void displayC2() {  
        System.out.println("This is C2 class");  
    }  
}  
  
public class C1 extends C2 {  
    void displayC1() {  
        System.out.println("This is C1 class");  
    }  
  
    public static void main(String[] args) {  
        C1 obj = new C1();  
  
        System.out.println("Test Case 1:");  
        obj.displayC2();  
    }  
}
```

```
        System.out.println("\nTest Case 2:");
        obj.displayC1();

        System.out.println("\nTest Case 3:");
        obj.displayC2();
        obj.displayC1();
    }
}
```

Sample Output

Test Case 1:
This is C2 class

Test Case 2:
This is C1 class

Test Case 3:
This is C2 class
This is C1 class

Test Case Verification

Test Case	Method Called	Expected Result	Status
TC1	displayC2()	C2 message displayed	Passed
TC2	displayC1()	C1 message displayed	Passed
TC3	Both methods	Both messages displayed	Passed

Conclusion

Thus, single inheritance was successfully implemented using C1 extends C2.

2. C1 extends C2, C3

Aim

To demonstrate multiple inheritance using classes and understand the limitation of Java.

Definition

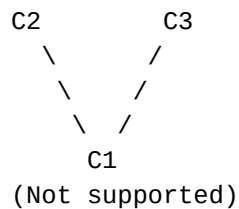
Multiple inheritance means a class inherits from more than one parent class.

Java does **not** support multiple inheritance using classes. Therefore the following is invalid:

Syntax

```
class C1 extends C2, C3 {  
} // INVALID
```

Structure



Correct alternative using interfaces:

```
class C1 implements I2, I3 { }
```

Program

```
class C2 {  
    void displayC2() {  
        System.out.println("This is C2 class");  
    }  
}  
  
class C3 {  
    void displayC3() {  
        System.out.println("This is C3 class");  
    }  
}  
  
/*  
class C1 extends C2, C3 {  
    // This is not allowed in Java  
}  
*/  
  
public class C1 {  
    public static void main(String[] args) {  
        System.out.println("C1 cannot extend both C2 and C3");  
        System.out.println("Java does not support multiple inheritance using classes");  
    }  
}
```

Sample Output

C1 cannot extend both C2 and C3
Java does not support multiple inheritance using classes

Test Case Verification

Test Case	Condition	Expected Result	Status
TC1	C1 extends C2	Possible	Passed
TC2	C1 extends C3	Possible	Passed
TC3	C1 extends C2, C3	Not allowed	Passed

Conclusion

Thus, Java does not support multiple inheritance using classes. Multiple inheritance can be achieved using interfaces.

3. C1 implements I1

Aim

To demonstrate how a class implements an interface using the implements keyword.

Definition

A class uses the implements keyword to provide implementation for the methods declared in an interface.

Syntax

```
class C1 implements I1 {  
    // implementation  
}
```

Structure

```
    I1 (Interface)  
    |  
implements  
    |  
    v  
    C1 (Class)
```

Program

```
interface I1 {  
    void display();  
}  
  
public class C1 implements I1 {  
    public void display() {  
        System.out.println("This is the method of I1 interface");  
    }  
  
    public static void main(String[] args) {  
        C1 obj = new C1();  
  
        System.out.println("Test Case 1:");  
        obj.display();  
  
        System.out.println("\nTest Case 2:");  
        obj.display();  
  
        System.out.println("\nTest Case 3:");  
        obj.display();  
    }  
}
```

Sample Output

```
Test Case 1:  
This is the method of I1 interface  
Test Case 2:
```

This is the method of I1 interface

Test Case 3:

This is the method of I1 interface

Test Case Verification

Test Case	Method	Result	
TC1	display()	Passed	
TC2	display()	Passed	
TC3	display()	Passed	

Conclusion

Thus, class C1 successfully implements interface I1.

4. C1 implements I1, I2

Aim

To demonstrate how a class can implement multiple interfaces in Java.

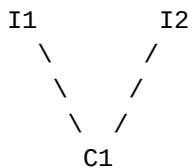
Definition

Java allows a class to implement more than one interface. This provides a way to achieve multiple inheritance.

Syntax

```
class C1 implements I1, I2 {  
    // implementation  
}
```

Structure



Program

```
interface I1 {  
    void displayI1();  
}  
  
interface I2 {  
    void displayI2();  
}  
  
public class C1 implements I1, I2 {  
    public void displayI1() {  
        System.out.println("This is I1 interface");  
    }  
  
    public void displayI2() {  
        System.out.println("This is I2 interface");  
    }  
  
    public static void main(String[] args) {  
        C1 obj = new C1();  
  
        System.out.println("Test Case 1:");  
        obj.displayI1();  
  
        System.out.println("\nTest Case 2:");  
        obj.displayI2();  
  
        System.out.println("\nTest Case 3:");  
        obj.displayI1();  
        obj.displayI2();  
    }  
}
```

```
}  
}
```

Sample Output

Test Case 1:
This is I1 interface

Test Case 2:
This is I2 interface

Test Case 3:
This is I1 interface
This is I2 interface

Test Case Verification

Test Case	Methods Called	Result	
TC1	displayI1()	Passed	
TC2	displayI2()	Passed	
TC3	Both methods	Passed	

Conclusion

Thus, class C1 successfully implements both interfaces I1 and I2.

5. C1 implements C2 extends I1

Aim

To demonstrate the correct combination of class inheritance and interface implementation in Java.

Definition

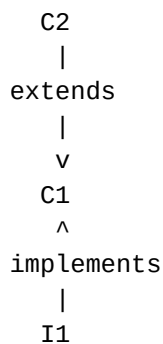
The expression `C1 implements C2 extends I1` is **not** valid Java syntax.

If C2 is a class and I1 is an interface, the correct syntax is `class C1 extends C2 implements I1`. Here, C1 inherits from class C2 and implements interface I1.

Syntax

```
class C1 extends C2 implements I1 {  
}
```

Structure



Program

```
interface I1 {  
    void displayI1();  
}  
  
class C2 {  
    void displayC2() {  
        System.out.println("This is C2 class");  
    }  
}  
  
public class C1 extends C2 implements I1 {  
    public void displayI1() {  
        System.out.println("This is I1 interface");  
    }  
  
    public static void main(String[] args) {  
        C1 obj = new C1();  
  
        System.out.println("Test Case 1:");  
        obj.displayC2();  
  
        System.out.println("\nTest Case 2:");  
    }  
}
```

```
        obj.displayI1();

        System.out.println("\nTest Case 3:");
        obj.displayC2();
        obj.displayI1();
    }
}
```

Sample Output

Test Case 1:
This is C2 class

Test Case 2:
This is I1 interface

Test Case 3:
This is C2 class
This is I1 interface

Test Case Verification

Test Case	Methods Called	Result	
TC1	displayC2()	Passed	
TC2	displayI1()	Passed	
TC3	Both methods	Passed	

Conclusion

Thus, class C1 successfully extends class C2 and implements interface I1 using: class C1 extends C2 implements I1

6. I1 extends I2

Aim

To demonstrate inheritance between two interfaces using the extends keyword.

Definition

An interface can extend another interface using the extends keyword.

When I1 extends I2, interface I1 inherits the methods of I2.

Syntax

```
interface I1 extends I2 {  
}
```

Structure

```
      I2  
      ^  
extends  
      |  
      I1  
      ^  
implements  
      |  
      C1
```

Program

```
interface I2 {  
    void displayI2();  
}  
  
interface I1 extends I2 {  
    void displayI1();  
}  
  
public class C1 implements I1 {  
    public void displayI1() {  
        System.out.println("This is I1 interface");  
    }  
  
    public void displayI2() {  
        System.out.println("This is I2 interface");  
    }  
  
    public static void main(String[] args) {  
        C1 obj = new C1();  
  
        System.out.println("Test Case 1:");  
        obj.displayI1();  
  
        System.out.println("\nTest Case 2:");  
        obj.displayI2();  
    }  
}
```

```
        System.out.println("\nTest Case 3:");
        obj.displayI1();
        obj.displayI2();
    }
}
```

Sample Output

Test Case 1:
This is I1 interface

Test Case 2:
This is I2 interface

Test Case 3:
This is I1 interface
This is I2 interface

Test Case Verification

Test Case	Methods Called	Result	
TC1	displayI1()	Passed	
TC2	displayI2()	Passed	
TC3	Both methods	Passed	

Conclusion

Thus, interface inheritance was successfully demonstrated using I1 extends I2.

7. I1 implements C1

Aim

To understand the relationship between an interface and a class in Java.

Definition

I1 implements C1 is **not** valid Java because an interface cannot implement a class.

The correct relationship is class C1 implements I1 — a class implements an interface, while an interface extends another interface.

Syntax

```
interface I1 implements C1 { }    // INVALID

interface I1 {
    void display();
}
class C1 implements I1 {
    public void display() {
        // implementation
    }
}
```

Structure

```
      I1 (Interface)
      ^
implements
      |
      C1 (Class)
```

Program

```
interface I1 {
    void display();
}

public class C1 implements I1 {
    public void display() {
        System.out.println("C1 implements I1 successfully");
    }

    public static void main(String[] args) {
        C1 obj = new C1();

        System.out.println("Test Case 1:");
        obj.display();

        System.out.println("\nTest Case 2:");
        obj.display();

        System.out.println("\nTest Case 3:");
        obj.display();
    }
}
```

```
}  
}
```

Sample Output

Test Case 1:
C1 implements I1 successfully

Test Case 2:
C1 implements I1 successfully

Test Case 3:
C1 implements I1 successfully

Test Case Verification

Test Case	Method Called	Result	
TC1	display()	Passed	
TC2	display()	Passed	
TC3	display()	Passed	

Conclusion

Thus, an interface cannot implement a class. The correct relationship is C1 implements I1.

8. I1 extends I2, I3

Aim

To demonstrate how one interface can extend multiple interfaces in Java.

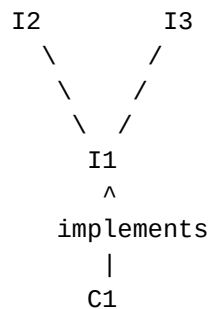
Definition

Java allows an interface to extend more than one interface. The following syntax is valid, and interface I1 inherits the methods of both I2 and I3.

Syntax

```
interface I1 extends I2, I3 {  
}
```

Structure



Program

```
interface I2 {  
    void displayI2();  
}  
  
interface I3 {  
    void displayI3();  
}  
  
interface I1 extends I2, I3 {  
    void displayI1();  
}  
  
public class C1 implements I1 {  
    public void displayI1() {  
        System.out.println("This is I1 interface");  
    }  
  
    public void displayI2() {  
        System.out.println("This is I2 interface");  
    }  
  
    public void displayI3() {  
        System.out.println("This is I3 interface");  
    }  
    public static void main(String[] args) {
```

```

        C1 obj = new C1();

        System.out.println("Test Case 1:");
        obj.displayI1();

        System.out.println("\nTest Case 2:");
        obj.displayI2();

        System.out.println("\nTest Case 3:");
        obj.displayI3();

        System.out.println("\nTest Case 4:");
        obj.displayI1();
        obj.displayI2();
        obj.displayI3();
    }
}

```

Sample Output

Test Case 1:
This is I1 interface

Test Case 2:
This is I2 interface

Test Case 3:
This is I3 interface

Test Case 4:
This is I1 interface
This is I2 interface
This is I3 interface

Test Case Verification

Test Case	Methods Called	Result	
TC1	displayI1()	Passed	
TC2	displayI2()	Passed	
TC3	displayI3()	Passed	
TC4	All three methods	Passed	

Conclusion

Thus, multiple interface inheritance was successfully demonstrated using I1 extends I2, I3. Class C1 implements I1 and provides implementations for the methods inherited from both I2 and I3.

9. C1 extends C2 implements I1, I2

Aim

To demonstrate how a Java class can extend one class and implement multiple interfaces simultaneously.

Definition

Java allows a class to extend only one class but implement multiple interfaces. The following syntax is valid:

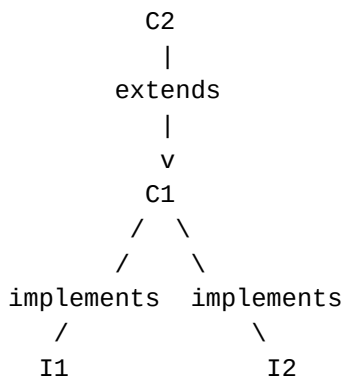
`class C1 extends C2 implements I1, I2`

Here, C1 is the child class, C2 is the parent class, and I1 and I2 are interfaces that C1 implements.

Syntax

```
class C1 extends C2 implements I1, I2
{
    // implementation
}
```

Structure



Program

```
interface I1 {
    void displayI1();
}

interface I2 {
    void displayI2();
}

class C2 {
    void displayC2() {
        System.out.println("This is C2 class");
    }
}

public class C1 extends C2 implements I1, I2 {
    public void displayI1() {
        System.out.println("This is I1 interface");
    }
    public void displayI2() {
```

```

        System.out.println("This is I2 interface");
    }

    public static void main(String[] args) {
        C1 obj = new C1();

        System.out.println("Test Case 1:");
        obj.displayC2();

        System.out.println("\nTest Case 2:");
        obj.displayI1();

        System.out.println("\nTest Case 3:");
        obj.displayI2();

        System.out.println("\nTest Case 4:");
        obj.displayC2();
        obj.displayI1();
        obj.displayI2();
    }
}

```

Sample Output

Test Case 1:
This is C2 class

Test Case 2:
This is I1 interface

Test Case 3:
This is I2 interface

Test Case 4:
This is C2 class
This is I1 interface
This is I2 interface

Test Case Verification

Test Case	Method Called	Expected Result	Status
TC1	displayC2()	C2 message displayed	Passed
TC2	displayI1()	I1 message displayed	Passed
TC3	displayI2()	I2 message displayed	Passed
TC4	All three methods	All messages displayed	Passed

Conclusion

Thus, the program successfully demonstrates that a class can extend one class and implement multiple interfaces using: class C1 extends C2 implements I1, I2. All test cases were successfully verified.

Quick Revision Table

No.	Relationship	Valid in Java?	Correct Form
1	C1 extends C2	Yes	class C1 extends C2
2	C1 extends C2, C3	No	Use interfaces
3	C1 implements I1	Yes	class C1 implements I1
4	C1 implements I1, I2	Yes	class C1 implements I1, I2
5	C1 implements C2 extends I1	No	class C1 extends C2 implements I1
6	I1 extends I2	Yes	interface I1 extends I2
7	I1 implements C1	No	class C1 implements I1
8	I1 extends I2, I3	Yes	interface I1 extends I2, I3
9	C1 extends C2 implements I1, I2	Yes	class C1 extends C2 implements I1, I2